Activity 7.1: Replicating Latin Squares

CO2 Emissions

> **Note**
>
> Submit via Gradescope, don't forget to assign pages to the questions in the outline!

Suppose researchers are interested in 2 gasoline blends (summer, winter), and an additive at 3 levels (2%, 3%, 4%). The response is CO emission under real driving conditions. We know there will be differences in CO emission among types of cars, as well as among driving conditions/routes.

a. Identify the following

- Response: ______________________________
- Factor(s) + Levels:

- Blocking factor 1: __________________________
- Blocking factor 2: __________________________

- How many rows and column blocking factor levels should be included in a Latin Square design for this study?

b. Sketch a possible study design for a standard unreplicated Latin Square (e.g., one "platform").

c. Sketch a possible study design for replicating the latin square by reusing cars (i.e., increase the number of routes to 12).

d. Write out a skeleton ANOVA for the replicated latin square in (4).